AHV Appliance Build Guide

ZTF-Orchestrator with baked ZeroTouch Framework and optional NKP framework and bundle preload

Field	Value
Repository	VirtuArchitect/ZTF-Orchestrator
Primary artifact	AHV-importable QCOW2 appliance
ZTF framework baked into app image	/opt/zerotouch-framework
NKP framework container path	/var/lib/ztf-orchestrator/nkp-zerotouch-framework
NKP bundle container path	/var/lib/ztf-orchestrator/bundles
Current ZTF-Orchestrator release	v1.8.5-rc1

Purpose. This guide describes how to build a portable ZTF-Orchestrator AHV appliance for connected staging or air-gapped Nutanix AHV environments. For this source tree, the current ZTF-Orchestrator release is v1.8.5-rc1. The current build process can bake the ZTF-Orchestrator application image, legacy ZeroTouch Framework v1.5.2, and NKP framework assets into the QCOW2 before transfer.

Build Host Requirements

- Use a connected Linux build host or GitHub Actions for the actual QCOW2 build.
- Install Git, QEMU, qemu-utils, and HashiCorp Packer on local Linux build hosts.
- Ensure outbound access to Ubuntu cloud images, GitHub, the ZeroTouch Framework repository, and the NKP framework repository when NKP bake-in is enabled.
- For air-gapped handoff, use an approved media or artifact transfer process for the finished QCOW2 and SHA256SUMS file.

```
sudo apt-get update
sudo apt-get install -y git qemu-system-x86 qemu-utils packer
```

Build Variables

Variable	Default / Example	Purpose
VERSION	v1.8.5-rc1 or main	Git ref staged into the appliance source checkout.
ZTF_ORCHESTRATOR_VERSION	v1.8.5-rc1 or latest	Container tag the appliance runs.
ZTF_BUILD_CONTAINER_IMAGE	true	Builds the app image inside the QCOW2.
ZTF_FRAMEWORK_REF	v1.5.2	Legacy ZeroTouch Framework ref baked into the app image.
ZTF_BAKE_NKP_FRAMEWORK	true	Clones or copies the NKP framework into the appliance.
ZTF_NKP_FRAMEWORK_REF	main	NKP framework ref to bake.
ZTF_NKP_BUNDLE_URLS	comma-separated URLs	Optional bundle downloads during connected staging builds.
QEMU_ACCELERATOR	kvm or tcg	Use kvm on Linux hosts with virtualization; use tcg otherwise.

Step-by-Step Local Build

1. Clone the repository on a connected Linux build host.
2. Optionally stage local NKP bundle artifacts before running Packer.
3. Optionally stage a reviewed local NKP framework checkout if you do not want Packer to clone it from Git.
4. Run the AHV QCOW2 build wrapper with ZTF and NKP bake-in enabled.
5. Verify the generated SHA256SUMS file.
6. Transfer the QCOW2 and checksum to the target AHV environment.

```
git clone https://github.com/VirtuArchitect/ZTF-Orchestrator.git
cd ZTF-Orchestrator
```

```
# Optional local bundle staging before build
cp /path/to/nkp-bundle-or-binary appliance/preload/bundles/

# Optional local NKP framework staging before build
rsync -a /path/to/nkp-zerotouch-framework/ appliance/preload/nkp-zerotouch-framework/

cd appliance
VERSION=v1.8.5-rc1 \
ZTF_ORCHESTRATOR_VERSION=v1.8.5-rc1 \
ZTF_BUILD_CONTAINER_IMAGE=true \
ZTF_PULL_CONTAINER_IMAGES=true \
ZTF_FRAMEWORK_REF=v1.5.2 \
ZTF_BAKE_NKP_FRAMEWORK=true \
ZTF_NKP_FRAMEWORK_REF=main \
QEMU_ACCELERATOR=kvm \
scripts/build-ahv-qcow2.sh

cd appliance/packer/output
sha256sum -c SHA256SUMS
```

GitHub Actions Build Path

- Open Actions > Build AHV Appliance Image in the repository.
- Select Run workflow.
- Use source_ref main or a release tag, image_version latest or the release tag, and qemu_accelerator tcg for GitHub-hosted runners.
- Leave build_container_image, pull_container_images, and bake_nkp_framework set to true for a portable appliance.
- Set nkp_bundle_urls only when bundles are available from approved connected staging URLs.
- Download the ztf-orchestrator-ahv-qcow2 artifact and verify SHA256SUMS before importing into AHV.

Workflow Input	Recommended Value
source_ref	main or release tag
image_version	latest or release tag
qemu_accelerator	tcg
build_container_image	true
pull_container_images	true
ztf_framework_ref	v1.5.2

bake_nkp_framework	true
nkp_framework_ref	main
nkp_bundle_urls	blank or approved comma-separated URLs

Import Into Nutanix AHV

7. Upload the QCOW2 into Prism Central or Prism Element image management.
8. Create a VM from the image with 2 vCPU, 4-8 GB RAM, 80-100 GB disk, and the management VLAN.
9. Provide cloud-init or your normal image process for administrator SSH access.
10. Boot the VM and wait for first boot to finish.
11. Open http://<appliance-ip>:5001 and sign in as admin using the generated password from service logs.

```
sudo systemctl status ztf-orchestrator-firstboot
sudo systemctl status ztf-orchestrator
sudo journalctl -u ztf-orchestrator -n 200 --no-pager
```

Post-Boot Validation

Check	Command / Path
Compose services	sudo docker compose -f appliance/docker-compose.appliance.yml ps
Application health	curl http://localhost:5001/health
Baked ZTF framework	ls -la /opt/zerotouch-framework inside the ztf-orchestrator container
Preloaded NKP framework	ls -la /var/lib/ztf-orchestrator/nkp-zerotouch-framework inside the container
Preloaded NKP bundles	ls -la /var/lib/ztf-orchestrator/bundles inside the container

```
cd /opt/ztf-orchestrator
sudo docker compose -f appliance/docker-compose.appliance.yml exec ztf-orchestrator ls -la
/opt/zerotouch-framework
```

```
sudo docker compose -f appliance/docker-compose.appliance.yml exec ztf-orchestrator ls -la
/var/lib/ztf-orchestrator/nkp-zerotouch-framework
sudo docker compose -f appliance/docker-compose.appliance.yml exec ztf-orchestrator ls -la
/var/lib/ztf-orchestrator/bundles
curl http://localhost:5001/health
```

Current operating baseline reviewed 2 November 2026: merged application a3c9c79; consolidated offline bundle v1.8.5-rc1-636f191. v1.8.4 remains Latest stable. Build/downloaded image provenance and target-appliance activation are separate checks.

After boot, verify the actual image and /health installed version tag, then use Start Here. Configure secure references and optional datacenter/cluster credential groups; organize saved YAML using the same group catalogue. Group labels never change runtime references or access permissions.

For guided upgrades, install the matching explicit host-service kit once and apply the reviewed Compose mounts during the initial cutover. Routine ZIP updates then use Appliance Ops > Updates. Preserve current backups, the previous image and rollback configuration. Completed stages are green; final health verification remains required.

NKP UI Registration

- Open NKP Framework in the ZTF-Orchestrator UI.
- Confirm the NKP framework path is /var/lib/ztf-orchestrator/nkp-zerotouch-framework.
- Register NKP bundle or binary paths under Binaries using /var/lib/ztf-orchestrator/bundles/<artifact>.
- Run Check CLI Compatibility before generating profiles or submitting NKP phases.
- Use only safe phases through the orchestrator unless a separately reviewed apply/destructive mode is added later.

> Security note: appliance runtime secrets are generated on first boot, not during image build. NKP preload mounts are read-only inside the app container. Do not commit NKP bundles or site-specific secrets to Git; use appliance/preload only as a local staging area.